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Ultimate chromatic polynomials

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Abstract

We outline an approach to enumeration problems which relies on the algebra of free abelian groups, giving as our main application a generalisation of the chromatic polynomial of a simple graph G. Our polynomial lies in the free abelian group generated by the poset $K(G)$ of contractions of G, and reduces to the classical case after a simple substitution. Its main properties may be stated in terms of an evaluation homomorphism, one for each positive integer t, which when applied to the polynomial yields an explicit list of the colourings of G with t colours. Considering posets larger than $K(G)$, and enriching the algebra accordingly, we extend the whole construction to incorporate the corresponding incidence Hopf algebras. The enriched polynomial reduces by a similar substitution to the umbral chromatic polynomial of G.

1. Introduction

Given a finite set S of combinatorially significant objects, the basic aim of enumeration theory is to investigate the cardinality $|S|$ of S. The most desirable solution is to obtain a closed formula, but it is often necessary to settle for a generating function or recursion relation, or for an expression in terms of other, and hopefully simpler, combinatorial quantities. We are particularly interested in the situation where S is the set of colourings of a simple graph G. In this case, the classical chromatic polynomial $\chi(G;x)$ (e.g. see [1]) provides complete information, its value at $x=t$ yielding $|S|$ for t colours, given any nonnegative integer t.

Amongst the most powerful and familiar techniques of enumeration, in increasing order of sophistication, are sieve methods, inclusion–exclusion principles, and Möbius inversion in posets. These proceed by a finite sequence of steps, each pair of which involves counting elements systematically but with repetitions, and

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overcompensating for the repetitions by subsequent subtractions. At each stage control over the error increases, until the final step counts the elements of the original set precisely. As first defined by Whitney [13], and reinterpreted by Rota [10], the chromatic polynomial fits naturally into this scheme.

The formulae produced by such techniques clearly always involve minus signs, and it is tempting to ask for an algebraic framework in which these signs are incorporated at as close a level as possible to the combinatorial data, rather than simply as operations on integers. In other words, we seek a signed analogue of the bijection approach to enumeration formulae [8]. We here propose to describe such a framework, which by construction is the simplest available, yet which offers an alternative perspective on several familiar results. It is set in the context of free abelian groups, enriched later to encompass free modules and Hopf algebras. We concentrate on applications to chromatic polynomials, since they illustrate well the general theoretical principles.

Many authors have introduced algebraic structure into enumeration theory, and inclusion-exclusion formulae have been a source of particular interest. They are often susceptible to symbolic description and manipulation (by processes such as the umbral calculus; e.g. see [9]), and these processes may be put on a sound footing by choosing the correct algebraic setting; two attractive examples are studied in [2]. Again, following the lead pioneered in [4], the second author has developed the study of antipodes in incidence Hopf algebras [11] as a contribution to the theory of Möbius inversion. We show here that our framework is not only consistent with his, but also sheds further light on its relevance.

By way of motivation, we remark that one obvious way to display the elements of a set S is as a list, so that solving the enumeration problem then consists of counting the elements in the resulting list, bearing in mind that the order in which they occur is immaterial. Therefore, we introduce $\mathbf{Z}\{S\}$, the *free abelian group on S*, whose elements consist of all formal finite linear combinations

$$\sum_i n_i x_i, \quad \text{where } n_i \in \mathbf{Z} \text{ and } x_i \in S.$$

In particular, $\sum_S x$ lies in $\mathbf{Z}\{S\}$, and consists of the unordered list of elements of S; it is therefore convenient to refer to this sum as the *list* of S, and abbreviate it to $\ell(S)$. Note that $\mathbf{Z}\{\emptyset\}$ is $\{0\}$.

There is a natural homomorphism $\theta : \mathbf{Z}\{S\} \to \mathbf{Z}$, which is defined by linearly extending the assignment $x \mapsto 1$ for all x in S. Clearly $\theta(\ell(S))$ is $|S|$, and our programme consists of developing enumeration techniques in the setting of $\mathbf{Z}\{S\}$, then recovering the classical formulae by the application of θ.

To apply such techniques to the chromatic theory of G, we work with the free abelian group on T^V, the set of functions from the vertices V to a set T of t colours. Our aim is to understand the subset $C(G;T)$ of T^V consisting of all colourings of G with T, by computing its list in $\mathbf{Z}\{T^V\}$. Our generalised chromatic polynomials are themselves defined in a suitable free abelian group, and $\ell(C(G;T))$ is obtained as the image of an evaluation homomorphism e_T. Since this process yields an explicit list of all the colourings of G, we refer to our polynomial as the ultimate chromatic polynomial of G, and write it as $\omega(G;X)$. Applying θ to the ultimate chromatic polynomial recovers the classical chromatic polynomial, and to $\ell(C(G;T))$ recovers the number of colourings with t colours.

In Section 2, we set the scene by describing, in our terms, a simple example of an inclusion–exclusion formula. We then extend the concepts so as to include Möbius inversion in posets. In fact, this extension is already part of the literature of Möbius algebras [3], but is usually presented with extra algebraic structure that we do not need, and with proofs that are derived from the standard ones, rather than as their precursors.

The main example is offered in Section 3, where we define the ultimate chromatic polynomial in terms of the poset of contractions of the graph G, and explain its main properties. We also discuss how the polynomial may be obtained from other posets constructed out of G.

In Section 4, we broaden our algebraic outlook, and replace the integers by a certain version of the incidence Hopf algebra of a given poset P. We take the free module generated by P over this algebra, and show how Möbius inversion may be defined in this setting, thereby making transparent a conjectured relationship in [11]. The procedure reduces to the free abelian group case by substituting the integer 1 for each interval of P.

Finally, in Section 5, we describe correspondingly enriched versions of the chromatic polynomial based on the respective posets of contractions, admissible partitions, and noncolour partitions. Not only do these generalise the ultimate chromatic polynomial but the latter two reduce to the umbral chromatic polynomial of [6] under suitable algebraic substitution.

Throughout, we use the convention that a variable of summation may be emphasised with an underdot.

2. Free abelian groups

As an introduction, we describe our extension of a well-known enumeration formula.

Suppose that a finite set S contains subsets $W_1, W_2, \ldots, W_n$, and write $N=\{1,\ldots,n\}$. For each subset J of N, we define W_J to be the subset of S consisting of all those elements which lie in W_i for each i in J, but in no others. Then the number $|W_J|$ is traditionally determined by an inclusion–exclusion principle.

Our formulation is as follows.

Proposition 2.1. *We have that*

$$\ell(W_J)=\sum_{J\subseteq \underset{\cdot}{K}}(-1)^{|K-J|}\ell\left(\bigcap_{i\in K} W_i\right)$$

in $\mathbf{Z}\{S\}$.

Proof. Consider x in S; it must lie in W_L for some subset L (possibly empty) of N. If L does not contain J, then x occurs on neither side of the equation, and if L is the same as J it occurs with coefficient 1 on both sides. On the other hand, if L properly contains J, then x occurs on the right-hand side with coefficient

$$\sum_{J \subseteq K \subseteq L} (-1)^{|K-J|} = (1-1)^{|L-J|},$$

which is zero as required. □

Applying θ immediately yields the standard formula

$$|W_J| = \sum_{J \subseteq K} (-1)^{|K-J|} \left| \bigcap_{i \in K} W_i \right|.$$

We now replace S by a poset P, which in all our examples will be locally finite (indeed finite), and will have a minimum element 0 and a maximum element 1. We consider $\mathbf{Z}\{P\}$ as before; with the list $\ell(P)$ it is natural to define two endomorphisms a and b by the rules

$$a(x) = \sum_{x \leqslant y} y \quad \text{and} \quad b(x) = \sum_{w \leqslant x} w, \tag{2.1}$$

for each x in P. They are extended to all of $\mathbf{Z}\{P\}$ by linearity, in the usual fashion, and their values may be referred to as the lists *above* and *below* x, respectively.

We wish to consider the process of Möbius inversion in P as stemming from two very simple identities in $\mathbf{Z}\{P\}$, which we prove below. In fact, these are well known in the literature since $\mathbf{Z}\{P\}$ is the basic additive structure underlying the Möbius algebra [3] of P. However, we do not require either the extra algebraic structure of the Möbius algebra or the standard formula for Möbius inversion in the incidence algebra of P; we emphasise that our aim is to demonstrate the ease of taking these identities as a starting point for the theory.

We require the integer valued *Möbius function*

$$m(x,y) = \sum_{\gamma} (-1)^{l(\gamma)}, \tag{2.2}$$

where the summation is taken over all chains $\gamma = \{x < \cdots < t < \cdots < y\}$ (not necessarily maximal) between x and y in P, and $l(\gamma)$ denotes the length. Thus, if $x > y$, the value of $m(x,y)$ is zero.

In what follows, it is convenient to write $s(\gamma)$ for the minimum element, or *source* of a chain γ, and $t(\gamma)$ for its maximum element, or *target*. We have the following proposition.

Proposition 2.2 (Möbius inversion). *The endomorphisms a and b are isomorphisms, with inverses given by*

$$a^{-1}(x)=\sum_{x\leqslant y} m(x,y)y \quad \text{and} \quad b^{-1}(x)=\sum_{w\leqslant x} m(w,x)w,$$

respectively.

Proof. To show that $a\cdot a^{-1}$ is the identity, we must verify that

$$x=\sum_{s(\gamma)=x}(-1)^{l(\gamma)}\sum_{t(\gamma)\leqslant z} z,$$

for each x in P. So consider the coefficient of z in the right-hand side, noting that if $z<x$ or $z=x$, it is 0 or 1, respectively. If $z>x$ it is

$$\sum_{s(\gamma)=x}(-1)^{l(\gamma)},$$

where the summation takes place over all chains with $t(\gamma)\leqslant z$. This is seen to be zero by pairing each chain γ whose target is less than z, with its counterpart obtained from γ by adjoining the interval $t(\gamma)<z$; their contributions cancel.

The same method shows that $a^{-1}\cdot a$ is the identity, and the proof for b^{-1} follows by dualisation of P. $\square$

To recover the standard form of Möbius inversion, we let $f\colon P\to A$ be a function into an abelian group A. This naturally extends to a homomorphism $\mathbf{Z}\{P\}\to A$, which we also denote by f. Then if g is the composition $f\cdot a$ or $f\cdot b$, so f is the composition $g\cdot a^{-1}$ or $g\cdot b^{-1}$, respectively. Applying Proposition 2.2 yields the familiar statements

$$g(x)=\sum_{x\leqslant y} f(y) \Leftrightarrow f(x)=\sum_{x\leqslant y} m(x,y)g(y), \tag{2.3}$$

$$g(x)=\sum_{w\leqslant x} f(w) \Leftrightarrow f(x)=\sum_{w\leqslant x} m(w,x)g(w), \tag{2.4}$$

in A. We see in Section 5 how these formulae are related to the Hopf algebra antipodes of [11]. For now, we conclude this section by explaining how a specific case may be used to rederive Proposition 2.1 above.

We work in the poset 2^N of subsets of the indexing set N, ordered by the inclusion relation. For two such subsets J and K, with $J\subseteq K$, the Möbius function $m(J,K)$ is easily computed to be $(-1)^{|K-J|}$.

Now define two homomorphisms $f,g\colon\mathbf{Z}\{2^N\}\to\mathbf{Z}\{S\}$ by specifying that

$$f(K)=\ell(W_K) \quad \text{and} \quad g(K)=\ell\left(\bigcap_{i\in K} W_i\right)$$

on generators. Since $\bigcap_{i\in J} W_i=\bigsqcup_{J\subseteq K} W_K$, it is clear that $g(J)=\sum_{J\subseteq K} f(K)$. Thus, by applying (2.3), we obtain $f(J)=\sum_{J\subseteq K} m(J,K)g(K)$, which is Proposition 2.1.

3. Chromatic polynomials

We now introduce the ultimate chromatic polynomial.

We are concerned solely with *simple* graphs $G=G(V,E)$, which have no loops or multiple edges. A colouring of G with t colours (or with the set T) is a function from the vertex set V into a set of colours $T=\{x_1,\ldots,x_t\}$, such that no element of the edge set E has the same colour assigned to both its vertices. Thus, every colouring f specifies a *colour partition* $\pi(f)$ of V into t blocks, each of which is necessarily independent. We think of $\pi(f)$ as an element of the poset $\Pi(V)$ of partitions of V, ordered by refinement. Note that the discrete partition V and the trivial partition $\{V\}$ are the elements 0 and 1 of $\Pi(V)$, respectively.

When Whitney introduced the chromatic polynomial $\chi(G;x)$, his proof of its properties [13] (and the motivation for his title) involved an application of the inclusion–exclusion principle to the poset of *bond closed* subsets of $E(G)$. Such a subset A is characterised by the fact that any edge of G not in A must have its vertices in distinct connected components of the subgraph $G(V,A)$. By assigning to A the partition π_A whose blocks are made up of the vertices of the connected components of $G(V,A)$, we may interpret this poset as the subposet $K(G)$ of *contractions* in $\Pi(V)$. A contraction of G is a partition of the vertices such that each block induces a connected subgraph of G.

In practice we deal with an alternative representation of $K(G)$, involving a suitably large arbitrary finite set X. For each contraction σ of G, we write X^σ for the set of functions $\sigma\to X$, which, by composition with the natural collapse $V\to\sigma$, may be construed as the set of functions $V\to X$ having $\pi(f)$ refined by σ. The sets X^σ, ordered by containment, clearly constitute a subposet of the dual of 2^{X^V}. This subposet is isomorphic to $K(G)$, and henceforth we identify the two. Its minimum element 0 is the set of all functions $V\to X$, and its maximum element 1 is the subset of all functions constant on connected components.

We may now make the following definition.

Definition 3.1. The *ultimate* chromatic polynomial of G is the element

$$\omega(G;X)=\sum_{s(\gamma)=0}(-1)^{l(\gamma)}X^{t(\gamma)},$$

in the free abelian group $\mathbf{Z}\{K(G)\}$.

Recalling the homomorphism a from Section 2, we see that $\omega(G;X)$ may be written as $a^{-1}(0)$. The observant reader will note that the ultimate chromatic polynomial is not actually a polynomial, since it exists in a structure for which no product has yet been defined. However, as suggested by the notation, an external product is available involving disjoint union of vertex sets: but we shall not pursue this matter here.

As our description of the properties of the ultimate chromatic polynomial will show, $\omega(G;X)$ reduces to $\chi(G;x)$ after replacing X^σ by $x^{|\sigma|}$ for each contraction σ.

This replacement defines an additive homomorphism $\theta : \mathbf{Z}\{K(G)\} \to \mathbf{Z}[x]$, closely related to the homomorphism θ of Section 1 by means of (3.1); basically, both are cardinality maps.

We require the *evaluation* map e_T, which is to be thought of as replacing the variable X in $\omega(G;X)$ by the list of colours $\ell(T)$. It is a homomorphism $\mathbf{Z}\{K(G)\} \to \mathbf{Z}\{T^V\}$, and is defined by the rule

$$e_T(X^\sigma) = \ell(T^\sigma),$$

for each contraction σ.

We are now able to describe the central property of the ultimate chromatic polynomial, whose image under e_T may be conveniently denoted by $\omega(G;T)$.

Proposition 3.2. *For any graph G and set of colours T, the list $\ell(C(G;T))$ of colourings of G is given by $\omega(G;T)$.*

Proof. To each f in T^V, we can associate the contraction $\kappa(f)$ obtained by splitting the blocks of $\pi(f)$ into connected components. Thus, there is a homomorphism $k : \mathbf{Z}\{K(G)\} \to \mathbf{Z}\{T^V\}$ defined on generators by $k(X^\sigma) = \sum_{\kappa(f)=\sigma} f$ and therefore satisfying

$$\sum_{\sigma \leqslant \rho} k(X^\rho) = \sum_{\sigma \leqslant \kappa(f)} f = e_T(X^\sigma),$$

for every contraction σ. Hence $e_T = k \cdot a$, so $k = e_T \cdot a^{-1}$ and

$$\omega(G;T) = e_T \cdot a^{-1}(0) = k(0) = \ell(C(G;T)),$$

as required. $\square$

This proof is, of course, an application of Proposition 2.2. As such it could be made more explicit by cancelling the chains in $K(G)$ directly, and so returning to the spirit of Whitney's original inclusion–exclusion version.

The best way to display the relationship between the ultimate and the classical polynomials is by means of the commutative square

$$\begin{array}{ccc} \mathbf{Z}\{K(G)\} & \xrightarrow{e_T} & \mathbf{Z}\{T^V\} \\ \theta \downarrow & & \downarrow \theta \\ \mathbf{Z}[x] & \xrightarrow{e_t} & \mathbf{Z} \end{array}, \tag{3.1}$$

where e_t denotes the substitution of x by t. Here, $\omega(G;X)$ maps to $\ell(C(G;T))$ horizontally and to $\chi(G;x)$ vertically; both then map in $\mathbf{Z}$ to the number of colourings of G by T.

It is of great interest to note that we may replace $K(G)$ in the above constructions by either of two other subposets of $\Pi(V)$. These are, respectively, the poset $A(G)$ of *admissible* partitions [6], each of whose blocks is either a singleton or contains an edge, and the poset $N(G)$ of *noncolour* partitions, at least one of whose blocks fails to be independent. These are related by the inclusions $K(G) \subseteq A(G) \subseteq N(G)$, and $\omega(G; X)$ may be defined by Definition 3.1 in any one of these. The only amendment required is the definition of a suitable version of $\kappa(f)$, as given in Section 5 below; we leave the details to the reader. It is interesting to speculate on the possible existence of other natural posets with the same property.

We conclude this section by outlining a more algorithmic procedure for computing the evaluation homomorphism, which again takes advantage of the setting of free abelian groups.

Given our set of colours $T = \{x_1, \ldots, x_t\}$, we may construct the free *monoid* F^+T on T, whose elements are words of the form $x_{i_1} x_{i_2} \cdots x_{i_n}$, multiplied by concatenation. The free abelian group $\mathbf{Z}\{F^+T\}$ may then be identified with the underlying additive group of the noncommutative polynomial algebra $\mathbf{Z}\langle T\rangle$ generated by T. Note that the subgroup generated by words of length one is exactly $\mathbf{Z}\{T\}$. Now suppose that the set of n vertices V is ordered, so that any partition π of V may be considered as having its ith block made up of integers $\pi_{i,1}, \pi_{i,2}, \ldots, \pi_{i,n_i}$, where $0 < \pi_{i,j} \leqslant n$. Then the set of functions T^V embeds in $\mathbf{Z}\langle T\rangle$ as the monomials of length n, according to the rule $f \mapsto f(1)f(2)\cdots f(n)$. This may be extended by linearity to a homomorphism ι defined on all of $\mathbf{Z}\{T^V\}$. Thus, we may compose e_T with ι, and ask for a description of the resulting evaluation map d_T.

We wish to replace the variable X in the ultimate chromatic polynomial by the list of colours $\ell(T)$, which already lies in $\mathbf{Z}\langle T\rangle$. Thus, to describe d_T we need to evaluate expressions of the form

$$(x_1 + x_2 + \cdots + x_t)^\sigma, \tag{3.2}$$

in $\mathbf{Z}\langle T\rangle$, for each contraction σ. This we do in two stages. First, we replace σ by its cardinality (or number of blocks) k, and expand into a sum of monomials of length k using the noncommutative product in $\mathbf{Z}\langle T^V\rangle$. Then we enlarge each such monomial $x_{r_1} x_{r_2} \cdots x_{r_k}$ to a monomial of length n by choosing x_{r_i} as the $\pi_{i,j}$th variable, for every i and j.

It is simple to check that this procedure does indeed describe $d_T : \mathbf{Z}\{K(G)\} \to \mathbf{Z}\langle T\rangle$, and therefore applies to the ultimate chromatic polynomial of a graph with ordered vertices to yield the list of colourings in $\mathbf{Z}\langle T\rangle$.

4. Enriching the algebra

In this section, we enrich the algebraic setting of our previous constructions by extending the ring of scalars from the integers to an incidence Hopf algebra. Our motivation is to establish links with the viewpoint of [11] and to make precise the relationship conjectured there between antipodes and the Möbius functions.

As before, we assume given a finite poset P, and we begin by describing a version of the incidence coalgebra [4] of P.

Let $I(P)$ denote the set of all intervals $[x, y]$ of P; they generate a polynomial algebra $\mathbf{Z}[I(P)]$, in which the multiplication is commutative by definition. We impose relations on this algebra by insisting that all trivial intervals of the form $[x, x]$ shall be equated with 1, and we label the resulting ring H_P. It is, of course, the polynomial algebra generated by all nontrivial intervals in P, but we prefer our original description since it simplifies some of our constructions, and also offers to knowledgeable reader the opportunity to proceed more delicately by localisation as in [11]. Note that by assigning the integer 1 to each interval, we obtain a homomorphism $\eta : H_P \to \mathbf{Z}$. For this, and any other function f defined on H_P, we abbreviate $f([x, y])$ to $f[x, y]$.

A crucial fact about H_P is that any chain γ in P defines an element $[\gamma]$ in H_P simply by juxtaposing the constituent intervals.

Now H_P may be given a coalgebra structure by means of the diagonal

$$\delta[x, z] = \sum_{x \leqslant y \leqslant z} [x, y] \otimes [y, z],$$

extended by linearity and multiplicativity. Thus, the first and last terms in this sum are $1 \otimes [x, z]$ and $[x, z] \otimes 1$, respectively. For H_P to be a Hopf algebra (see [12]), it therefore remains only to construct an antipode; in other words an automorphism which is a convolution inverse to the identity map.

Proposition 4.1. *An antipode μ for H_P is defined by*

$$\mu[x, y] = \sum_{\gamma} (-1)^{l(\gamma)} [\gamma],$$

where the summation is taken over all chains γ between x and y in P.

Proof. For each generator $[x, y]$ of H_P, it suffices to establish that $(\mu \otimes 1) \cdot \delta[x, z]$ multiplies out to zero. This expression is given by

$$\sum_{s(\gamma) = x} (-1)^{l(\gamma)} [\gamma][t(\gamma), z],$$

where the summation takes place over all chains with $t(\gamma) \leqslant z$. This is seen to be zero by pairing each term $[\gamma][t(\gamma), z]$, where $t(\gamma) < z$, with its counterpart $[\gamma']$, where γ' is obtained from γ by adjoining the interval $t(\gamma) < z$; their contributions cancel. The other properties of an antipode now follow readily. □

Given this antipode, H_P is very closely related to the free incidence Hopf algebras of [11]. Furthermore, the antipode reduces to the Möbius function of (2.2) under η.

An immediate corollary of the above proof (and a standard property of the antipode) concerns the left H_P linear *shear* endomorphism c of $H_P \otimes H_P$ defined by

$$c(g \otimes h) = \sum gh' \otimes h'',$$

where $\delta(h) = \sum h' \otimes h''$.

Corollary 4.2. *The endomorphism c is an isomorphism, with inverse given by*

$$c^{-1}(g \otimes h) = \sum g\mu(h') \otimes h''.$$

This property may alternatively be written as

$$1 \otimes [w,z] = \sum_{s(\gamma)=w} (-1)^{l(\gamma)}[\gamma] \sum_{t(\gamma) \leqslant y \leqslant z} [t(\gamma), y] \otimes [y, z], \tag{4.1}$$

for each interval $[w,z]$ in P, thereby expressing the fact that $c \cdot c^{-1}$ is the identity.

We now are in a position to enrich our previous constructions as promised. Our basic generalisation of the free abelian group on P is $H_P\{P\}$, the *free H_P module on P*, whose elements consist of all formal finite linear combinations

$$\sum_i h_i x_i, \quad \text{where } h_i \in H_P \text{ and } x_i \in P.$$

Of course $\mathbf{Z}\{P\}$ is a subgroup of $H_P\{P\}$, onto which η is a projection.

Observe that $H_P\{P\}$ admits two natural H_P linear monomorphisms i_a and i_b into $H_P \otimes H_P$, defined by $x \mapsto 1 \otimes [x, 1]$ and $x \mapsto 1 \otimes [0, x]$, respectively. Furthermore, the shear map c respects both of these embeddings, and restricts to endomorphisms A and B, respectively, of $H_P\{P\}$ defined by

$$A(x) = \sum_{x \leqslant y} [x, y]y \quad \text{and} \quad B(x) = \sum_{w \leqslant x} [w, x]w,$$

for each x in P. These clearly reduce under η to the lists a and b of (2.1).

Proposition 4.3 (enriched Möbius inversion). *The endomorphisms A and B are isomorphisms, with inverses given by*

$$A^{-1}(x) = \sum_{x \leqslant y} \mu[x, y]y \quad \textit{and} \quad B^{-1}(x) = \sum_{w \leqslant x} \mu[w, x]w,$$

respectively.

Proof. Restrict Corollary 4.2 to the images of i_a and i_b, respectively. □

Thus are the antipode and the Möbius function identified. Note from (4.1) that this proof exactly parallels that of Proposition 2.2, to which it reduces under η.

The standard forms of enriched Möbius inversion now follow automatically, the data being a function $f: P \to M$ for any H_P module M, and the output being the relationship between f and the compositions $f \cdot A$ and $f \cdot B$, generalising (2.3) and (2.4).

5. Enriched chromatic polynomials

Throughout this section, we let P stand for any one of the posets $K(G)$, $A(G)$ or $N(G)$ defined in Section 3, specialising wherever necessary. We continue to interpret P as a subposet of the dual of 2^{X^V}, given by finite set X.

We wish to explore ways of enriching the ultimate chromatic polynomial in accordance with the algebra of Section 4, and to describe connections with the umbral chromatic polynomial of [6]. The reader may, with some justification, ask why this is necessary, since the ultimate polynomial provides a complete description of colourings, and is therefore superior to the umbral version (a fact which surprises us, as we had always assumed that the poset $A(G)$, rather than $K(G)$, was required to extract the umbral information). However, we have found the enriched setting to be unavoidable in making these connections algebraically precise; and it certainly offers one of the best motivations for the process of umbral substitution that we have yet seen. Also, by virtue of Section 4, it justifies our long-standing belief in a direct link between the umbral chromatic polynomial and the work of [11].

We therefore make the following definition.

Definition 5.1. The ultimate chromatic polynomial of G *enriched in* P is the element

$$\omega^P(G;X) = \sum_{s(\gamma)=0} (-1)^{l(\gamma)}[\gamma]X^{t(\gamma)},$$

in the free H_P module $H_P\{P\}$.

Recalling the homomorphism A from Section 4, we note that $\omega^P(G;X)$ may be written as $A^{-1}(0)$. Thus, whichever P is chosen, $\omega^P(G;X)$ reduces to $\omega(G;X)$ under η.

Given a set of colours T, we need to describe a new evaluation map e_T^P. For this purpose, we associate to each f in T^V a partition $\lambda(f)$ as follows. If P is $K(G)$, we choose $\lambda(f)$ to be $\kappa(f)$ as in Proposition 3.2; if P is $A(G)$ we choose it to be $\alpha(f)$, obtained by splitting into singletons those blocks of $\pi(f)$ which contain no edges; and if P is $N(G)$ we choose it to be $\nu(f)$, obtained by splitting the blocks of $\pi(f)$ into singletons only when none of them contain an edge. Then e_T^P is the H_P linear homomorphism $H\{P\}\to H_P\{T^V\}$ defined by the rule

$$e_T^P(X^\sigma) = \sum_{T^\sigma} [X^\sigma, X^{\lambda(f)}]f.$$

We use this map to extend Proposition 3.2, writing $e_T^P(\omega^P(G;X))$ as $\omega^P(G;T)$.

Proposition 5.2. *For any graph G and set of colours T, the list $l(C(G;T))$ of G is given by $\omega^P(G;T)$.*

Proof. To each f in T^V, we may associate the partition $\lambda(f)$ in P. Thus, there is a homomorphism $k^P: H_P\{P\}\to H_P\{T^V\}$ defined on generators by $k^P(X^\sigma)=\sum_{\lambda(f)=\sigma} f$ and therefore satisfying

$$e_T^P(X^\sigma) = \sum_{\rho\geqslant\sigma} [X^\sigma, X^\rho]k^P(X^\rho),$$

for every σ in P. Hence $e_T^P = k^P \cdot A$, so $k^P = e_T^P \cdot A^{-1}$ and

$$\omega^P(G;T) = e_T^P \cdot A^{-1}(0) = k^P(0) = \ell(C(G;T)),$$

as required. $\square$

This proof obviously reduces to that of Proposition 3.2 under η, and could be made more explicit by cancelling the chains in P directly.

For each of the three choices of P, the polynomials $\omega^P(G;X)$ will in general be different, as the reader may readily check by considering the graph on four vertices with a single edge. However, when P is either $A(G)$ or $N(G)$ they share a common reduction to the umbral chromatic polynomial, by analogy with the way in which $\omega(G;X)$ reduces to the classical case.

To explain this, recall from [6] the polynomial algebra Φ, generated over $\mathbf{Z}$ by the sequence of elements $\phi_1, \phi_2, \ldots$, and consider the ring homomorphism $H_P \to \Phi$ defined on generators by $[X^\pi, X^\sigma] \mapsto \tau(\sigma/\pi)$. Here $\tau(\rho)$ denotes the *type* of any partition ρ, given by the monomial $\prod_{i>0} \phi_i^{n_i}$, where n_i is the number of blocks of ρ having $i+1$ elements. It will cause no confusion if we also use τ to denote the homomorphism.

We may extend τ in two different ways. Firstly, we obtain $H_P\{P\} \to \Phi[x]$ by insisting that X^σ maps to $x^{|\sigma|}$, and secondly, we obtain $H_P\{T^V\} \to \Phi$ by insisting that f maps to $\tau(\pi(f))$. In order to establish the analogue of (3.1), we label both of these homomorphisms as θ^P; they are linked by the umbral evaluation map [6] $e_{t\phi} : \Phi[x] \to \Phi$, which is defined on each x^n to be $\sum_{T^n} \tau(\pi(f))$, and is extended by linearity over Φ.

Proposition 5.3. *Let P be $A(G)$ or $N(G)$. Then there is a commutative square of homomorphisms*

$$\begin{array}{ccc} H_P\{P\} & \xrightarrow{e_T^P} & H_P\{T^V\} \\ {\scriptstyle\theta^P}\downarrow & & \downarrow{\scriptstyle\theta^P} \\ \Phi[x] & \xrightarrow{e_{t\phi}} & \Phi \end{array} \tag{5.1}$$

in which $\omega^P(G;X)$ maps to $\ell(C(G;T))$ horizontally, and to the umbral chromatic polynomial $\chi^\phi(G;x)$ vertically; both then map in Φ to the list $\sum_{C(G;T)} \tau(\pi(f))$.

Proof. To begin, we chase X^σ around both sides of the square. For the results to agree (so that the diagram commutes), we see that we must have

$$\tau(\pi(f)/\sigma) = \tau(\lambda(f)/\sigma) \cdot \tau(\pi(f)/\lambda(f)), \tag{5.2}$$

for each f such that $\pi(f)$ is refined by σ. This is true in general only if the amalgamations creating $\pi(f)$ out of $\lambda(f)$ involve distinct elements from the amalgamations creating $\lambda(f)$ out of $\sigma(f)$, a condition which is readily verified for $\alpha(f)$ in $A(G)$ and $\nu(f)$ in $N(G)$, but which fails for $\kappa(f)$ in $K(G)$.

Having established that the diagram commutes, chasing $\omega^P(G;X)$ around it becomes a formality. The only point to note is that when P is $A(G)$, it maps vertically down to $\chi^{\Phi}(G; x)$ according to the original definition [6] of the latter. That the same is true when P is $N(G)$ follows either by an explicit chain cancelling argument, or else by applying the umbral interpolation principal [7] to the result. $\square$

The conditions (5.2) has already been shown to be relevant to the understanding of the role of $A(G)$, albeit less directly, in [5]. Moreover, an alternative and more mysterious description of $e_{t\phi}$ in terms of the umbral integer $t\phi$ has regularly been used in the literature; this can now be simply explained by appeal to the evaluation procedure of (3.2). We map each x^n to the noncommutative product $(x_1 + \cdots + x_t)^n$, expand, impose commutativity, and replace each x_i^n with ϕ_{n-1}, thus measuring the type of the colouring. We leave the details to the interested reader.

To conclude, we remark that (3.1) and (5.1) are compatible in the sense that we may map the first into the second. Thus, if we apply our homomorphism η to the top row, and the corresponding ring homomorphism of [6], which replaces each ϕ_i with 1, to the bottom row, we obtain a commutative cube. This cube captures the essence of our work.

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